Department	General Studies	Major									
Course Name	Mathematics	Course Code	MATH 101								
Prerequisites	---	Credit Hours CRH	3			CTH		4			
			L	3	P	0	T	1			

Course Description:

The course includes the topics needed for the specialization such as: Sets, Number systems, Polynomials, Matrices and Determinants, linear and quadratic equations, system of linear equations and finally Area and Volumes computations.

General Objective:

This course provides an introduction of principle topics that enable the student to understand the specialized mathematics courses.

Detailed Objectives:

Trainee will be able to:

1-	Be familiar with elementary nations of Sets and perform simple logical operations on them
2-	Know the number systems and manipulate numbers.
3-	Be familiar with algebra of Polynomials. Know how to factorize Polynomials and simplify Algebraic fractions
4-	Deal with Matrices and Determinants and know how to use them.
5-	Solve first order equations, second order equations, and system of linear equations in two and three variables
6-	Be familiar with how to calculate areas, perimeters and volumes of planar and solid geometrical shapes

Detailed of Theoretical Contents			
Hours	**Contents**		**Assessment Tools**
10	**Sets:** • Definition of a set and its characteristics . • Universal set, empty set. • Subsets of a set, equality of two sets. • Operations on sets (union, intersection, difference, symmetric difference). • Complement of a set and De Morgan's Theorem. • Numbers sets and their symbols.		Oral questions Written questions Self-test Black-board
	Textbook	1	**Classic Set Theory : For Guided Independent Study**, D.C Goldrei, Taylor & Francis Group, 1St Edition, 28 July 1996.
		2	**Real Analysis : An Introduction to the Theory of Real Functions and Integration**, Jewgeni H. Dshalalow, Taylor & Francis Group, 1St Edition, 28 September 2000.

Detailed of Theoretical Contents			
Hours	**Contents**		**Assessment Tools**
10	**Mathematical operations on number sets:** • Mathematical Operations on numbers sets (addition, subtraction, multiplication, division) • Priority of mathematical operations in algebraic expressions • Rational numbers and operations on them • Decimal approximation of the real numbers		Oral questions Written questions Self-test Black-board
	Textbook	1	**Engineering Mathematics,** KA. Stroud, Macmillan Press, seventh edition 2013
		2	**Real Analysis : An Introduction to the Theory of Real Functions and Integration**, Jewgeni H. Dshalalow, Taylor & Francis Group, 1St Edition, 28 September 2000
10	**Polynomials:** • Definition of polynomials (coefficients and degree). • Operations on polynomials (addition, subtraction, multiplication, and long division). • Factorization of quadratic polynomials SEP. • Simplification of algebraic fractions.		Oral questions Written questions Self-test Black-board
	Textbook	1	**Advanced Calculus An introduction to Modern Analysis**, Voxman. Boca Raton, Taylor & Francis Group, 1St Edition, 19 October 2017, Chapter 1.
		2	**Basic Mathematics,** H. Kruglak & J. T. Moore, SCHAUM Outlines, McGraw-Hill
10	**Matrices and Determinants:** • Concept of matrices and their types. • Mathematical operations on matrices. • Calculation of determinants (2x2 and 3x3). • Calculation of inverse of matrices (2x2).		Oral questions Written questions Self-test Black-board
	Textbook	1	**Abstract Algebra An Inquiry Based Approach**, Jonathan k. Hodge, Taylor & Francis Group, 1St Edition, 21 December 2013
		2	**Engineering Mathematics,** KA. Stroud, Macmillan Press, seventh Edition 2013
12	**Equations:** • Solving first order equations in one variable. • Solving Quadratic Equations Using Discriminants. • Solving system of linear equations using Substitution Method. • Solving system of linear equations using Cramer Methods .		Oral questions Written questions Self-test Black-board
	Textbook	1	**Essential Mathematics for Engineers**, W.J.R.H Pooler, 1St Edition, 2011, Bookboon,

Detailed of Theoretical Contents				
Hours	**Contents**			**Assessment Tools**
		2	**Difference Equations Theory , Application and Advanced Topics**, 3 d Edition, Ronald E. Mickens, Taylor & Francis Group, 3rd Edition, 6 March 2015	
12	**Area and Volumes:** • Area and perimeter (square, triangle, circle, rectangle, trapezoidal) • Volumes (cubic, cylinder, cone, sphere)			Oral questions Written questions Self-test Black-board
	Textbook	1	**Essential Mathematics for Engineers**, W.J.R.H Pooler, 1St Edition, 2011, Bookboon,	